Time to get cosmic. There are more stars in the universe than grains of sand on any beach, more stars than seconds have passed since Earth formed, more stars than words and sounds ever uttered by all the humans who ever lived.

Want a sweeping view of the past? Our unfolding cosmic perspective takes you there. Light takes time to reach Earth's observatories from the depths of space, and so you see objects and phenomena not as they are but as they once were, back almost to the beginning of time itself. Within that horizon of reckoning, cosmic evolution unfolds continuously, in full view.

Want to know what we're made of? Again, the cosmic perspective offers a bigger answer than you might expect. The chemical elements of the universe are forged in the fires of high-mass stars that end their lives in titanic explosions, enriching their host galaxies with the chemical arsenal of life as we know it. The result? The four most common, chemically active elements in the universe—hydrogen, oxygen, carbon, and nitrogen—are the four most common elements of life on Earth, with carbon serving as the foundation of biochemistry.

We do not simply live in this universe. The universe lives within us.

That being said, we may not even be of this Earth. Several separate lines of research, when considered together, have forced investigators to reassess who we think we are and where we think we came from. As we've already seen, when a large asteroid strikes a planet, the surrounding areas can recoil from the impact energy, catapulting rocks into space. From there, they can travel to—and land on—other planetary surfaces. Second, microorganisms can be hardy. Extremophiles on Earth can survive wide ranges of temperature, pressure, and radiation encountered during space travel. If the rocky ejecta from an impact hails from a planet with life, then microscopic fauna could have stowed away in the rocks' nooks and crannies. Third, recent evidence suggests that shortly after the formation of our solar system, Mars was wet, and perhaps fertile, even before Earth was.

Collectively, these findings tell us it's conceivable that life began on Mars and later seeded life on Earth, a process known as panspermia. So all Earthlings might—just might—be descendants of Martians.

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Again and again across the centuries, cosmic discoveries have demoted our self-image. Earth was once assumed to be astronomically